DATA 202 — Quiz 3

ANSWER KEY — Version A

Question 1 (SLO 04A)

Table	Column	Answer
`members`	`member_id`	Primary key
`members`	`plan`	Foreign key -> plans
`plans`	`plan`	Primary key
`checkins`	`member_id`	Foreign key -> members (it repeats: one row per member per month)
`checkins`	`week_1`	Neither (a measurement, not an identifier)

Question 2 (SLO 04B)

how=	rows
`"inner"`	5
`"left"`	6
`"right"`	7
`"outer"`	8

M002 and M003 have 2 check-in rows each and M004 has 1, all matching a member, so inner = 5. Left (keep all check-ins) adds M006 (no member row) -> 6. Right (keep all members) adds M001 and M005 (no check-ins) -> 7. Outer adds all three -> 8.

Question 3 (SLO 04C)

```
checkins_long = pd.melt(
    checkins,
    id_vars=["member_id", "month"],
    value_vars=["week_1", "week_2", "week_3", "week_4"],
    var_name="week",
    value_name="visits",
)
```

member_id	month	week	visits
M002	Sep	week_1	6
M002	Sep	week_2	5
M002	Sep	week_3	6
M002	Sep	week_4	2
M002	Oct	week_1	6

member_id	month	week	visits
M002	Oct	week_2	1
M002	Oct	week_3	2
M002	Oct	week_4	4

8 rows: 2 wide rows x 4 week columns. Row order may differ (pd.melt actually lists all week_1 rows first, then week_2, …); any order is fine.

Question 4 (SLO 05A)

member	visits_per_week	calories (hundreds)	closest
M11	1	1	A
M12	1	3	A
M13	1	2	A
M14	4	3	B
M15	5	4	B
M16	6	2	B

New centroid A = (1, 2), new centroid B = (5, 3) – the mean visits_per_week and mean calories of each centroid's members.

Assign: each member goes to the centroid at the smallest straight-line distance. Update: each centroid moves to the average of its members.

Question 5 (SLO 05B)

k = 3. Inertia: the elbow is at 3 – it falls steeply from k = 1 to 3 (852 → 436 → 165), then only slowly (139, 116, 96, …). Silhouette: highest at k = 3 (0.61). Both agree. (Choosing k = 8 just because its inertia is lowest is wrong: inertia always drops as k grows.)

Question 6 (SLO 05C)

```
PCA().fit(X_scaled).explained_variance_ratio_.round(3)
# [0.617  0.307  0.070  0.006]
```

2 components: 0.617 + 0.307 = 0.924, so they keep **92.4%** (PC1 alone is only 61.7%).

Question 7 (SLO 03A) — Retake from Quiz 2

```
members["plan"] = (
    members["plan"]
    .str.strip()
    .str.lower()
    .str.replace(r"^prem$", "premium", regex=True)
)
```

member_id	plan (raw)	plan (after)
M001	`Basic`	`basic`
M002	`BASIC`	`basic`
M003	`Prem`	`premium`
M004	`basic`	`basic`
M005	`Student - Monthly`	`student - monthly`

The `^...$` anchors mean only an *exact* `"prem"` is replaced. Nothing fixes the spaces in `"student - monthly"`, so M005 would still not match `plans`.

Question 8 (SLO 03B) — Retake from Quiz 2

Answer:

```
result = (
    members.groupby("plan")["calories"]
    .mean()
    .reset_index()
    .sort_values("calories", ascending=False)
)
```

Question 9 (SLO 03C) — Retake from Quiz 2

Question	Answer
What's the distribution of `calories` across all members?	Histogram – x=calories
How do `avg_minutes` and `calories` relate to each other?	Scatter – x=avg_minutes, y=calories
How does the total number of `visits` change from `week_1` to `week_4` (using `checkins_long`)?	Line – x=week, y=sum(visits)
How does average `cardio_pct` compare across `plan` values?	Bar – x=plan, y=avg(cardio_pct)